Module-3

Laplace Transform

Let $F(t)$ be a function of t defined for all $t \geq 0$, then the Laplace Transform of $F(t)$ is defined as $L(F(t))$, which is given by,

$$L(F(t)) = \int_0^\infty e^{-pt} F(t)\, dt = f(p)$$

Provided that the integral exist, p is a parameter which may be real or complex. The function $f(p)$ is called image of $F(t)$.

Laplace Transform of Elementary Functions:

1. $L(1) = \frac{1}{p}$, $p > 0$

2. $L(t) = \frac{1}{p^2}$, $p > 0$

3. $L(t^n) = \frac{n!}{p^{n+1}}$, $p > 0$, n is a +ve integer

4. $L(t^n) = \frac{\lfloor n+1}{p^{n+1}}$, $p > 0$, $n > -1$

5. $L(e^{at}) = \frac{1}{p-a}$, $p > a$

6. $L(e^{-at}) = \frac{1}{p+a}$

7. $L(\sin at) = \frac{a}{p^2+a^2}$, $p > 0$

8. $L(\cos at) = \frac{p}{p^2+a^2}$, $p > 0$

9. $L(\sinh at) = \frac{a}{p^2-a^2}$, $p > |a|$

10. $L(\cosh at) = \frac{p}{p^2-a^2}$, $p > |a|$

Linearity Property:

If C_1 & C_2 are constants and f & g are functions of t, then

$$L(C_1 f(t) + C_2 g(t)) = C_1 L(f(t)) + C_2 L(g(t))$$

Transformations of Discontinuous Functions:

1. First translation or first shifting property,

   If $L(F(t)) = f(p)$ then $L(e^{at} F(t)) = f(p-a)$.

2. Second translation property or Heaviside's shifting theorem,

   If $L(F(t)) = f(p)$ & $G(t) = \begin{cases} F(t-a), & t \geq a \\ 0, & t < a \end{cases}$

   then

   $$L(G(t)) = e^{-ap} f(p).$$

3. Change of scale property,

   If $L(F(t)) = f(p)$ then $L(F(at)) = \frac{1}{a} f\left(\frac{p}{a}\right)$.

Q.1) Find the Laplace Transform of,
$7e^{2t} + 9e^{-2t} + 5\cos t + 7t^3 + 5\sin 3t + 2$.

Soln:

Let, $F(t) = 7e^{2t} + 9e^{-2t} + 5\cos t + 7t^3 + 5\sin 3t + 2$

Now,

$$L(F(t)) = L(7e^{2t} + 9e^{-2t} + 5\cos t + 7t^3 + 5\sin 3t + 2)$$

$$= 7L(e^{2t}) + 9L(e^{-2t}) + 5L(\cos t) + 7L(t^3) + 5L(\sin 3t) + 2L(1)$$

$$= \frac{7}{p-2} + \frac{9}{p+2} + \frac{5p}{p^2+1} + \frac{7\times 3!}{p^{3+1}} + \frac{5\times 3}{p^2+9} + \frac{2}{p}$$

$$= \frac{7}{p-2} + \frac{9}{p+2} + \frac{5p}{p^2+1} + \frac{42}{p^4} + \frac{15}{p^2+9} + \frac{2}{p}.$$

Q.2) Find Laplace transform of (i) $\sin 2t . \cos 3t$
(ii) $\sin^3 2t$

Sol^n:

(i) $F(t) = \sin 2t . \cos 3t$

$$\Rightarrow F(t) = \frac{1}{2}[2 \sin 2t . \cos 3t]$$

$$= \frac{1}{2}[\sin(2t+3t) + \sin(2t-3t)]$$

$$= \frac{1}{2}[\sin 5t + \sin(-t)]$$

$$\Rightarrow F(t) = \frac{1}{2}(\sin 5t - \sin t)$$

Now,

$$L(F(t)) = \frac{1}{2}[L(\sin 5t) - L(\sin t)]$$

$$= \frac{1}{2}\left[\frac{5}{p^2+25} - \frac{1}{p^2+1}\right]$$

$$\therefore L(F(t)) = \frac{5}{2(p^2+25)} - \frac{1}{2(p^2+1)}.$$

(ii) $F(t) = \sin^3 2t$

We have, $\sin 3t = 3\sin t - 4\sin^3 t$

$\Rightarrow \sin 6t = 3\sin 2t - 4\sin^3 2t$

$\Rightarrow \sin^3 2t = \frac{3}{4}\sin 2t - \frac{1}{4}\sin 6t$

Now,

$$L(F(t)) = L(\sin^3 2t) = \frac{3}{4}L(\sin 2t) - \frac{1}{4}L(\sin 6t)$$

$$= \frac{3}{2\cancel{4}}\left[\frac{\cancel{2}}{p^2+4}\right] - \frac{1}{\cancel{4}^2}\left[\frac{\cancel{6}^3}{p^2+36}\right]$$

$$= \frac{3}{2(p^2+4)} - \frac{3}{2(p^2+36)}$$

$$= \frac{3}{2}\left[\frac{\cancel{p^2}+36-\cancel{p^2}-4}{(p^2+4)(p^2+36)}\right]$$

$$= \frac{3 \times \cancel{32}^{16}}{\cancel{2}(p^2+4)(p^2+36)}$$

$$\therefore L(F(t)) = \frac{48}{(p^2+4)(p^2+36)}.$$

Q.3) Find the Laplace transform of $e^t t^{-1/2}$.

Sol^n:

~~Let, $F(t) = e^t t^{-1/2}$~~

First translation property,

If $L(F(t)) = f(p)$ then

$$L(e^{at} F(t)) = f(p-a)$$

For $e^t t^{-1/2}$, $F(t) = t^{-1/2}$

$$f(p) = L(F(t)) = L(t^{-1/2}) = \frac{\lfloor(-1/2)+1}{p^{-1/2+1}} = \frac{\lfloor 1/2}{p^{1/2}} = \frac{\sqrt{\pi}}{\sqrt{p}}$$

$$f(p) = \sqrt{\frac{\pi}{p}}$$

$$\Rightarrow f(p-a) = f(p-1) = \sqrt{\frac{\pi}{p-1}}$$

$$\therefore L(e^t t^{-1/2}) = \sqrt{\frac{\pi}{p-1}}.$$

$$\int uv\,dx = u\int v\,dx - \int\left(u' \int v\,dx\right)dx$$

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Q.4) Find the Laplace transform of

$$F(t) = \begin{cases} 1, & 0 \le t < 1 \\ +t, & 1 \le t < 2 \\ t^2, & 2 \le t < \infty \end{cases}$$

Soln:

$$L(F(t)) = \int_0^\infty e^{-pt} F(t)\,dt$$

$$= \int_0^1 e^{-pt}.1\,dt + \int_1^2 e^{-pt}.t\,dt + \int_2^\infty e^{-pt}t^2\,dt$$

$$= \left[\frac{e^{-pt}}{-p}\right]_0^1 + \left\{t\int e^{-pt}dt - \int\left(1\int e^{-pt}dt\right)dt\right\}_1^2$$

$$+ \left\{t^2\int e^{-pt}dt - \int\left(2t\int e^{-pt}dt\right)dt\right\}_2^\infty$$

$$= \left[\frac{e^{-pt}}{-p}\right]_0^1 + \left[\frac{t\,e^{-pt}}{-p} - \frac{e^{-pt}}{p^2}\right]_1^2 + \left[\frac{t^2e^{-pt}}{-p} - 2\int\frac{te^{-pt}}{-p}dt\right]_2^\infty$$

$$= \left[\frac{e^{-pt}}{-p}\right]_0^1 + \left[\frac{t\,e^{-pt}}{-p} - \frac{e^{-pt}}{p^2}\right]_1^2 + \left[\frac{t^2e^{-pt}}{-p} + \frac{2}{p}\int te^{-pt}dt\right]_2^\infty$$

$$= \left[\frac{e^{-pt}}{-p}\right]_0^1 + \left[\frac{t\,e^{-pt}}{-p} - \frac{e^{-pt}}{p^2}\right]_1^2 + \left\{\frac{t^2e^{-pt}}{-p} + \frac{2}{p}\left[\frac{te^{-pt}}{-p} - \frac{e^{-pt}}{p^2}\right]\right\}_2^\infty$$

$$\boxed{e^{-\infty} = 0}$$

$$= \frac{-1}{p}\left[e^{-p} - 1\right] + \left[\frac{2e^{-2p}}{-p} - \frac{e^{-2p}}{p^2} + \frac{e^{-p}}{p} + \frac{e^{-p}}{p^2}\right]$$

$$+ \left[\frac{4e^{-\infty}}{p} - \frac{4e^{-2p}}{p^2} - \frac{2e^{-2p}}{p^3}\right]$$

$$= \frac{-e^{-p}}{p} + \frac{1}{p} - \frac{2e^{-2p}}{p} - \frac{e^{-2p}}{p^2} + \frac{e^{-p}}{p} + \frac{e^{-p}}{p^2} + \frac{4e^{-2p}}{p}$$

$$+ \frac{4e^{-2p}}{p^2} + \frac{2e^{-2p}}{p^3}$$

$$\therefore L(F(t)) = \frac{1}{p} + \frac{2e^{-2p}}{p} + \frac{3e^{-2p}}{p^2} + \frac{e^{-p}}{p^2} + \frac{2e^{-2p}}{p^3}.$$

Q.5) Given that $L(\cos^2 t) = \frac{p^2+2}{p(p^2+4)}$. Find $L(\cos^2 at)$.

Soln:

Let, $F(t) = \cos^2 t$

$\Rightarrow F(at) = \cos^2 at$

$$L(F(at)) = \frac{1}{a} f\left(\frac{p}{a}\right)$$

$f(p) = L(F(t))$

$$\Rightarrow f(p) = \frac{p^2+2}{p(p^2+4)}$$

$$\Rightarrow f\left(\frac{p}{a}\right) = \frac{(p/a)^2 + 2}{(p/a)(p^2/a^2+4)} = \frac{a^3(p^2+2a^2)}{a^2[p(p^2+4a^2)]}$$

$$\therefore L(\cos^2 at) = \frac{1}{a}\left[\frac{a(p^2+2a^2)}{p(p^2+4a^2)}\right] = \frac{p^2+2a^2}{p(p^2+4a^2)}.$$

Functions of Exponential Order

A function $F(t)$ is said to be of exponential order as $t \to \infty$ if there exist a constant M and b, and a fixed value t_0 & t such that,

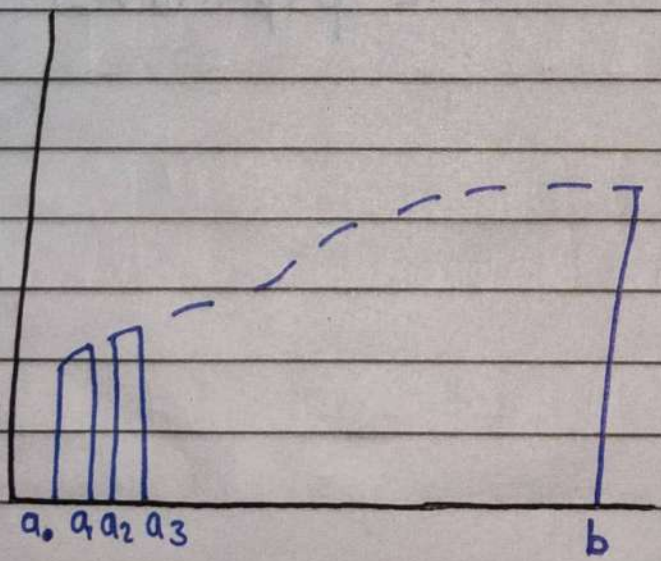
$$|F(t)| < Me^{bt} \quad , t \geq t_0 \, , t \to \infty$$

We also write,

$F(t) = O(e^{bt})$, $t \to \infty$ to mean that $F(t)$ is of Exponential order.

Function of class 'A'

A function which is sectionaly or piece-wise continuous over every finite interval in the range $t \geq 0$ and is of exponential order as $t \to \infty$ is termed as a function of class 'A'.



Existence Theorem

If a function $F(t)$ is of class 'A' then $L(F(t))$ exists.

Q.1) Prove that t^n is a function of exponential order as $t \to \infty$.

Sol^n:

Let, $F(t) = t^n$

$$\lim_{t\to\infty} e^{-bt} \cdot t^n = \lim_{t\to\infty} \frac{t^n}{e^{bt}} = 0 \text{ (exists)} \quad \left[e^{\infty} = \infty, \frac{1}{\infty} = 0 \right]$$

$$\left[e^{-bt} F(t) < M \right]$$

So, t^n is a function of Exponential Order.

Q.2) Show that e^{t^2} is not of exponential order as $t \to \infty$.

Soln:

$$\lim_{t\to\infty} e^{-bt} \cdot e^{t^2} = \lim_{t\to\infty} e^{t^2-bt} = \lim_{t\to\infty} e^{t(t-b)}$$

If $t = b$,

$$\lim_{t\to\infty} e^{0} = 1$$

If $b < t$,

$$\lim_{t\to\infty} e^{t(t-b)} = \infty$$

If $b > t$,

$$\lim_{t\to\infty} e^{t(t-b)} = 0$$

$\therefore$ For $b < t$,

$$\lim_{t\to\infty} e^{-bt} \cdot e^{t^2}$$ does not exist.

$\therefore$ It is not of exponential order.

#① Laplace Transform of Derivative of Function.

$$L(F'(t)) = P\,L(F(t)) - F(0) = P\,f(P) - f(0)$$

② Laplace Transform of Derivatives of nth order.

$$L(F^n(t)) = P^n\,L(F(t)) - P^{n-1}\,F(0) - P^{n-2}\,F'(0) - P^{n-3}\,F''(0) \ldots\ldots F^{n-1}(0).$$

③ Laplace of Function Multiplied by t^n.

If $L(F(t)) = f(P)$ then

$$L(t^n\,F(t)) = (-1)^n \frac{d^n}{dP^n}\,f(P) = (-1)^n \frac{d^n}{dP^n}\,L(F(t))$$

④ If $L(F(t)) = f(P)$ then

$$L\left(\int_0^t F(t)\,dt\right) = \frac{f(P)}{P}.$$

⑤ Division by t.

If $L(F(t)) = f(P)$ then

$$L\left(\frac{F(t)}{t}\right) = \int_P^\infty f(P)\,dP.$$

Q.1) If $F(t) = \frac{e^{at} - \cos bt}{t}$, find $L(F(t))$.

Soln:

Let, $F(t) = \frac{G(t)}{t} = \frac{e^{at} - \cos bt}{t}$

$G(t) = e^{at} - \cos bt$

$$L(F(t)) = L\left(\frac{G(t)}{t}\right) = \int_p^\infty g(p)\, dp$$

$[g(p) = L(G(t))]$

Now,

$g(p) = L(G(t)) = L(e^{at} - \cos bt)$

$\Rightarrow g(p) = L(e^{at}) - L(\cos bt)$

$\Rightarrow g(p) = \frac{1}{p-a} - \frac{p}{p^2+b^2}$

$$L(F(t)) = L\left(\frac{G(t)}{t}\right) = \int_p^\infty \left[\frac{1}{p-a} - \frac{p}{p^2+b^2}\right] dp$$

$$= \int_p^\infty \frac{dp}{(p-a)} - \int_p^\infty \frac{p}{(p^2+b^2)}\, dp$$

$$= \left[\log(p-a)\right]_p^\infty - \frac{1}{2}\int_{(p^2+b^2)}^\infty \frac{dx}{x}$$

$[\text{let } x = p^2+b^2 \Rightarrow dx = 2p\, dp]$

$$= \left[\log (p-a)\right]_p^{\infty} - \frac{1}{2}\left[\log x\right]_{(p^2+b^2)}^{\infty}$$

$$= \left[\log (p-a)\right]_p^{\infty} - \frac{1}{2}\left[\log (p^2+b^2)\right]_p^{\infty}$$

$$= \left[\log (p-a) - \log \sqrt{p^2+b^2}\right]_p^{\infty}$$

$$= \left[\log \left(\frac{p-a}{\sqrt{p^2+b^2}}\right)\right]_p^{\infty}$$

$$= \left[\log \left(\frac{\cancel{p}(1-a/p)}{\cancel{p}\sqrt{1+b^2/p^2}}\right)\right]_p^{\infty}$$

$$= \log \left(\frac{1-a/\infty}{\sqrt{1+b^2/\infty}}\right) - \log \left(\frac{1-a/p}{\sqrt{1+b^2/p^2}}\right)$$

$$= \cancel{\log (1)}^{0} - \log \left(\frac{p-a}{\sqrt{p^2+b^2}}\right)$$

$$= -\log \left(\frac{p-a}{\sqrt{p^2+b^2}}\right)$$

$$= \log \left(\frac{\sqrt{p^2+b^2}}{p-a}\right).$$

Q.2) If $L(t \sin \omega t) = \frac{2\omega p}{(p^2+\omega^2)^2}$, then evaluate $L(2\cos \omega t - \omega t \sin \omega t)$.

Soln:

$$L(2\cos \omega t - \omega t \sin \omega t)$$
$$= 2L(\cos \omega t) - \omega L(t \sin \omega t)$$
$$= \frac{2p}{p^2+\omega^2} - \omega\left(\frac{2\omega p}{(p^2+\omega^2)^2}\right)$$
$$= \frac{2p}{(p^2+\omega^2)} - \frac{2\omega^2 p}{(p^2+\omega^2)^2}$$
$$= \frac{2p^3 + 2\omega^2 p - 2\omega^2 p}{(p^2+\omega^2)^2}$$
$$= \frac{2p^3}{(p^2+\omega^2)^2}.$$

Q.3) Find Laplace transform of $t \sin^2 3t$.

Sol^n:

We have, $L(t^n F(t)) = (-1)^n \frac{d^n}{dp^n} L(F(t))$

Here, $F(t) = \sin^2 3t$ & $n = 1$

$$\Rightarrow L(\sin^2 3t) = L\left(\frac{1 - \cos 6t}{2}\right)$$

$$= \frac{1}{2}\left[L(1) - L(\cos 6t)\right]$$

$$= \frac{1}{2}\left[\frac{1}{p} - \frac{p}{p^2+36}\right]$$

Then,

$$L(t \sin^2 3t) = (-1)\frac{d}{dp} L(\sin^2 3t)$$

$$= -\frac{d}{dp}\left[\frac{1}{2p} - \frac{p}{2(p^2+36)}\right]$$

$$= -\left[\frac{-1}{2p^2} - \frac{(p^2+36).1 - p(2p)}{2(p^2+36)^2}\right]$$

$$= \frac{1}{2p^2} + \frac{36 - p^2}{2(p^2+36)^2}$$

$$= \frac{(p^2+36)^2 + 36p^2 - p^4}{2p^2(p^2+36)^2}$$

$$= \frac{\cancel{p^4} + 72p^2 + (36)^2 + 36p^2 \cancel{- p^4}}{2p^2(p^2+36)^2}$$

$$= \frac{108p^2 + (36)^2}{2p^2(p^2+36)^2}$$

$$= \frac{\overset{54}{\cancel{108}}(p^2+12)}{\cancel{2}p^2(p^2+36)^2}$$

$$\therefore L(t \sin^2 3t) = \frac{54(p^2+12)}{p^2(p^2+36)^2}.$$

Q.4) If $L\left(2\sqrt{\frac{t}{\pi}}\right) = \frac{1}{p^{3/2}}$. Show that $L\left(\frac{1}{\sqrt{\pi t}}\right) = \frac{1}{\sqrt{p}}$

Soln:

$$\because \frac{1}{t}\sqrt{\frac{t}{\pi}} = \frac{1}{\sqrt{\pi t}}$$

$$\therefore L\left(\frac{1}{\sqrt{\pi t}}\right) = L\left(\frac{\sqrt{t/\pi}}{t}\right)$$

We have,

$$L\left(\frac{F(t)}{t}\right) = \int_p^\infty f(p)\,dp = \int_p^\infty L(F(t))\,dp$$

Let, $F(t) = \sqrt{\frac{t}{\pi}}$

$$\frac{F(t)}{t} = \frac{\sqrt{t/\pi}}{t} = \frac{1}{\sqrt{\pi t}}$$

$$L\left(\frac{1}{\sqrt{\pi t}}\right) = L\left(\frac{\sqrt{t/\pi}}{t}\right)$$

$$= \int_p^\infty L\left(\sqrt{\frac{t}{\pi}}\right) dp$$

$$= \int_p^\infty \frac{1}{2p^{3/2}}\,dp$$

$$\left[L\left(2\sqrt{\frac{t}{\pi}}\right) = \frac{1}{p^{3/2}}\right]$$

$$\Rightarrow L\left(\sqrt{\frac{t}{\pi}}\right) = \frac{1}{2p^{3/2}}$$

$$= -\frac{1}{2}\left[\frac{2}{p^{1/2}}\right]_p^\infty$$

$$= -\left[\frac{1}{\sqrt{p}}\right]_p^\infty$$

$$= -\left[\frac{1}{\infty} - \frac{1}{\sqrt{p}}\right]$$

$$= -\left[0 - \frac{1}{\sqrt{p}}\right]$$

$$= \frac{1}{\sqrt{p}}$$

Thus, $L\left(\frac{1}{\sqrt{\pi t}}\right) = \frac{1}{\sqrt{p}}$ proved.

Q.5) Find the value of $\int_0^\infty \frac{e^{-t}\sin^2 t}{t}\,dt$.

Sol^n:

We have,

$$f(p) = L(F(t)) = \int_0^\infty e^{-pt} F(t)\,dt$$

Comparing with the given integral,

$$p = 1, \quad F(t) = \frac{\sin^2 t}{t}$$

$$\int_0^\infty \frac{e^{-t}\sin^2 t}{t}\,dt = L\left(\frac{\sin^2 t}{t}\right) = f(1)$$

So, $L\left(\frac{\sin^2 t}{t}\right) = \int_p^\infty L(\sin^2 t)\,dp$ $\left[L\left(\frac{F(t)}{t}\right) = \int_p^\infty L(F(t))\,dp\right]$

Now,

$$L(\sin^2 t) = L\left(\frac{1-\cos 2t}{2}\right)$$

$$= \frac{1}{2}\left[L(1) - L(\cos 2t)\right]$$

$$= \frac{1}{2}\left[\frac{1}{p} - \frac{p}{p^2+4}\right]$$

So, $L\left(\frac{\sin^2 t}{t}\right) = \frac{1}{2}\int_p^\infty \left[\frac{1}{p} - \frac{p}{p^2+4}\right] dp$

$$= \frac{1}{2}\int_p^\infty \frac{1}{p}\, dp - \frac{1}{2}\int_p^\infty \frac{p}{p^2+4}\, dp$$

Let, $p^2+4 = x \Rightarrow 2p\, dp = dx$

$$\Rightarrow p\, dp = \frac{dx}{2}$$

$$L\left(\frac{\sin^2 t}{t}\right) = \frac{1}{2}\int_p^\infty \frac{1}{p}\, dp - \frac{1}{2}\int_{p^2+4}^\infty \frac{1}{x}\cdot\frac{dx}{2}$$

$$= \frac{1}{2}\left[\log p\right]_p^\infty - \frac{1}{4}\left[\log x\right]_{p^2+4}^\infty$$

$$= \frac{1}{2}\left[\log p\right]_p^\infty - \frac{1}{4}\left[\log (p^2+4)\right]_p^\infty$$

$$= \frac{1}{2}\left[\log p - \log (p^2+4)^{1/2}\right]_p^\infty$$

$$= \frac{1}{2}\left[\log\left(\frac{p}{\sqrt{p^2+4}}\right)\right]_p^\infty$$

$$= \frac{1}{2}\left[\log\left(\frac{p}{\sqrt{p^2+4}}\right)\right]_p^\infty$$

$$= \frac{1}{2}\left[\log\left(\frac{\cancel{p}}{\cancel{p}\sqrt{1+4/p^2}}\right)\right]_p^\infty$$

$$= \frac{1}{2}\left[\log\left(\frac{1}{\sqrt{1+4/p^2}}\right)\right]_p^\infty$$

$$= \frac{1}{2}\left[\log\left(\frac{1}{\sqrt{1+4/\infty}}\right) - \log\left(\frac{1}{\sqrt{1+4/p^2}}\right)\right]$$

$$= \frac{1}{2}\left[\cancel{\log(1)}^{\to 0} - \log\left(\frac{p}{\sqrt{p^2+4}}\right)\right]$$

$$= \frac{-1}{2}\log\left(\frac{p}{\sqrt{p^2+4}}\right)$$

So, $L\left(\frac{\sin^2 t}{t}\right) = \frac{1}{2}\log\left(\frac{\sqrt{p^2+4}}{p}\right) = f(p)$

For $\int_0^\infty \frac{e^{-t}\sin^2 t}{t}\,dt$, $p = 1$

$\therefore \int_0^\infty \frac{e^{-t}\sin^2 t}{t}\,dt = f(1) = \frac{1}{2}\log(\sqrt{5})$.

Q.6) Find the laplace transform of $\int_0^t e^{-t}\cos t\,dt$.

Soln:

$$L\left(\int_0^t e^{-t}\cos t\,dt\right) = \frac{f(p)}{p}$$

$$\left[L\left(\int_0^t F(t)\,dt\right) = \frac{f(p)}{p}\right]$$

Now, $f(p) = L(e^{-t}\cos t)$

[If $L(F(t)) = f(p)$
then $L(e^{at}F(t)) = f(p-a)$]

$$L(e^{-t}\cos t) = f(p+1) \quad [\because a = -1]$$

So, $f(p) = L(\cos t) = \frac{p}{p^2+1}$

$$\Rightarrow f(p+1) = \frac{(p+1)}{(p+1)^2+1}$$

$$\therefore L\left(\int_0^t e^{-t}\cos t\,dt\right) = \frac{L(e^{-t}\cos t)}{p}$$

$$= \frac{(p+1)}{p[(p+1)^2+1]}.$$

Q.7) Find laplace transform of $\int_0^t \frac{\sin t}{t}\, dt$.

Sol^n:

We have, $L\left(\int_0^t F(t)\,dt\right) = \frac{f(p)}{p} = \frac{L(F(t))}{p}$

So, $L\left(\int_0^t \frac{\sin t}{t}\right) = \frac{L(\sin t/t)}{p}$

Now,

$$L\left(\frac{\sin t}{t}\right) = \int_p^\infty L(\sin t)\, dp$$

$$\left[L\left(\frac{F(t)}{t}\right) = \int_p^\infty f(p)\, dp = \int_p^\infty L(F(t))\, dp\right]$$

$$L\left(\frac{\sin t}{t}\right) = \int_p^\infty \frac{1}{p^2+1}\, dp$$

$$= \left[\tan^{-1} p\right]_p^\infty$$

$$= \tan^{-1}\infty - \tan^{-1} p$$

$$= \frac{\pi}{2} - \tan^{-1} p$$

$$= \cot^{-1} p$$

$\therefore L\left(\int_0^t \frac{\sin t}{t}\, dt\right) = \frac{\cot^{-1} p}{p}$.

Q.8) Find laplace transform of $t^2 e^t \sin 4t$.

Soln:

We have, $L(t^n F(t)) = (-1)^n \frac{d^n}{dp^n} f(p) = (-1)^n \frac{d^n}{dp^n} L(F(t))$

So, $L(t^2 e^t \sin 4t) = (-1)^2 \frac{d^2}{dp^2} L(e^t \sin 4t)$

$L(e^t \sin 4t) = f(p-1)$

$f(p) = L(\sin 4t) = \frac{4}{p^2+16}$

$\Rightarrow f(p-1) = \frac{4}{(p-1)^2+16}$

$\Rightarrow L(e^t \sin 4t) = \frac{4}{(p-1)^2+16}$

$\Rightarrow L(t^2 e^t \sin 4t) = \frac{d^2}{dp^2}\left[\frac{4}{(p-1)^2+16}\right]$

$\Rightarrow L(t^2 e^t \sin 4t) = \frac{d^2}{dp^2}\left[\frac{4}{p^2-2p+17}\right]$

Now, $\frac{d}{dp}\left[\frac{4}{p^2-2p+17}\right] = \frac{-4(2p-2)}{(p^2-2p+17)^2}$

$$\frac{d}{dp}\left[\frac{-4(2p-2)}{(p^2-2p+17)^2}\right]$$

$$= -8\,\frac{d}{dp}\left[\frac{p-1}{(p^2-2p+17)^2}\right]$$

$$= -8\left[\frac{(p^2-2p+17)^2 \cdot 1 - 2(p-1)\cdot(2p-2)(p^2-2p+17)}{(p^2-2p+17)^4}\right]$$

$$= -8\,\cancel{(p^2-2p+17)}\left[\frac{(p^2-2p+17)-4(p-1)^2}{(p^2-2p+17)^{\cancel{4}\,3}}\right]$$

$$= -8\left[\frac{p^2-2p+17-4p^2+8p-4}{(p^2-2p+17)^3}\right]$$

$$= -\frac{8(-3p^2+6p+13)}{(p^2-2p+17)^3}$$

$$\therefore L(t^2 e^t \sin 4t) = \frac{8(3p^2-6p-13)}{(p^2-2p+17)^3}.$$

Periodic Function

Let $F(t)$ is a periodic function of period $T > 0$, then,

$$L(F(t)) = \frac{\int_0^T e^{-pt} F(t)\, dt}{1 - e^{-pT}}$$

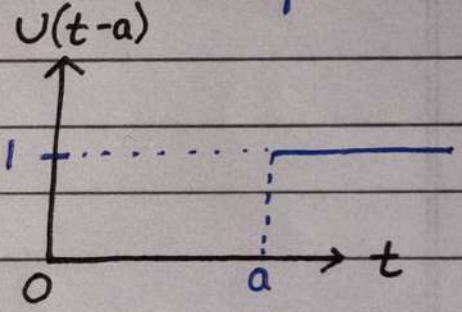
Unit Step Function

Unit step function is defined as,

$$U(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$$

It is also called ~~Heaviri~~ Heaviside Step Function.

$$L(U(t-a)) = \frac{e^{-ap}}{p}.$$



Second Shifting Theorem for Unit Step Function

If $L(F(t)) = f(p)$ then,

$$L(F(t-a).U(t-a)) = e^{-ap}.f(p).$$

General Formula:

1. If $$F(t) = \begin{cases} g(t), & 0 \le t < a \\ h(t), & t \ge a \end{cases}$$

then,

$$F(t) = g(t)[1 - U(t-a)] + h(t)\,U(t-a).$$

2. If $$F(t) = \begin{cases} g(t), & 0 \le t < a \\ j(t), & a \le t < b \\ h(t), & t \ge b \end{cases}$$

then,

$$F(t) = g(t)[1 - U(t-a)] + j(t)[U(t-a) - U(t-b)] + h(t)\,U(t-b).$$

Q.1) Using unit step function. Find laplace transform of:

(i) $(t-1)^2 U(t-1)$ (ii) $\sin t\, U(t-\pi)$

Soln:

(i) $L((t-1)^2 U(t-1)) = L(F(t-1) U(t-1)) = e^{-p} f(p)$

$F(t-1) = (t-1)^2$

$F(t) = t^2$

$a = 1$

Now,

$f(p) = L(F(t))$

$= L(t^2)$

$= \frac{2!}{p^3}$

$= \frac{2}{p^3}$

$\therefore L((t-1)^2 U(t-1)) = e^{-p} f(p) = \frac{2e^{-p}}{p^3}$.

(ii) We have, $\sin t \cdot U(t-\pi)$

$\sin t = \sin(\pi - t)$

$\Rightarrow -\sin(t-\pi) = \sin t$

$L(\sin t\, U(t-\pi)) = -L(\sin(t-\pi)\, U(t-\pi))$

From second shifting theorem, we have,

$L(F(t-a)\, U(t-a)) = e^{-ap} \cdot f(p)$

So, $L(\sin t\, U(t-\pi))$

$= -L(\sin(t-\pi)\, U(t-\pi))$

$= -e^{-\pi p} \cdot f(p)$

$f(p) = L(F(t)) = L(\sin t) = \dfrac{1}{p^2+1}$

$\therefore L(\sin t\, U(t-\pi)) = \dfrac{-e^{-\pi p}}{p^2+1}$.

Q.2) Find the laplace transform of the rectified semi-wave function defined by,

$$f(t) = \begin{cases} \sin \omega t, & 0 < t \leq \pi/\omega \\ 0, & \pi/\omega < t < 2\pi/\omega \end{cases}$$

Soln:

We Know,

$$L(f(t)) = \frac{\int_0^T e^{-pt} f(t)\, dt}{1 - e^{-pT}}$$

Here, $T = \frac{2\pi}{\omega}$

$$\Rightarrow L(f(t)) = \frac{\int_0^{2\pi/\omega} e^{-pt} f(t)\, dt}{1 - e^{-\frac{2\pi p}{\omega}}}$$

$$= \frac{\int_0^{\pi/\omega} e^{-pt} \sin \omega t}{1 - e^{-\frac{2\pi p}{\omega}}}$$

$$= \frac{\left[\frac{e^{-pt}}{p^2+\omega^2}\left(-p \sin \omega t - \omega \cos \omega t\right)\right]_0^{\pi/\omega}}{1 - e^{-\frac{2\pi p}{\omega}}}$$

$$\left[\int e^{ax} \sin(bx+c) = \frac{e^{ax}}{a^2+b^2}\left(a \sin bx - b \cos bx\right)\right]$$

$$= \frac{1}{p^2+\omega^2}\left[e^{-\frac{p\pi}{\omega}}\left(-p\sin\left(\omega.\frac{\pi}{\omega}\right) - \omega\cos\left(\omega.\frac{\pi}{\omega}\right)\right) - e^{0}\left(-p\sin 0 - \omega\cos 0\right)\right]}{1-e^{-\frac{2\pi p}{\omega}}}$$

$$= \frac{\omega.e^{-\frac{p\pi}{\omega}} + \omega}{(p^2+\omega^2)\left(1-e^{-\frac{2\pi p}{\omega}}\right)}$$

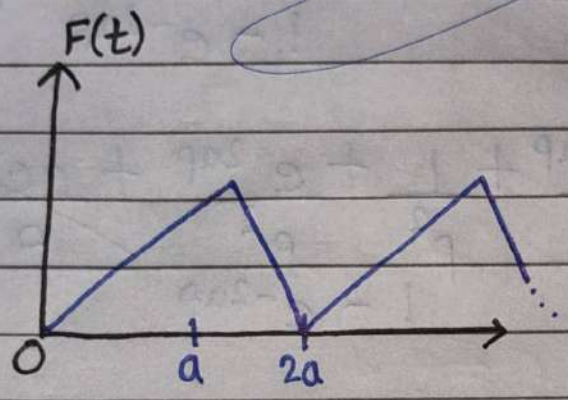
$$= \frac{\omega\left(1+e^{-\frac{p\pi}{\omega}}\right)}{(p^2+\omega^2)\left(1-e^{-\frac{\pi p}{\omega}}\right)\left(1+e^{-\frac{\pi p}{\omega}}\right)}$$

$$= \frac{\omega}{(p^2+\omega^2)\left(1-e^{-\frac{\pi p}{\omega}}\right)}.$$

Q.3) Draw the graph of the following function and find its laplace transform.

$$F(t) = \begin{cases} t, & 0 < t \le a \\ 2a - t, & a < t \le 2a \end{cases}$$

Soln:



Here, $T = 2a$

$$L(F(t)) = \frac{\int_0^{2a} e^{-pt} F(t)\, dt}{1 - e^{-2ap}}$$

$$= \frac{\int_0^a e^{-pt} \cdot t\, dt + \int_a^{2a} e^{-pt}(2a - t)\, dt}{1 - e^{-2ap}}$$

$$\left[\int e^x \cdot x\, dx = x e^x - \int e^x\right]$$

Now, $\int e^{-pt} t\, dt = -\frac{t e^{-pt}}{p} + \int \frac{e^{-pt}}{p} = -\frac{t e^{-pt}}{p} - \frac{e^{-pt}}{p^2}$

Also, $\int e^{-pt}(2a - t)\, dt = -\frac{(2a-t)e^{-pt}}{p} - \int \frac{e^{-pt}}{p}$

$$= -\frac{(2a-t)e^{-pt}}{p} + \frac{e^{-pt}}{p^2}$$

So, $L(f(t)) = \dfrac{\left[\dfrac{-te^{-pt}}{p} - \dfrac{e^{-pt}}{p^2}\right]_0^a + \left[\dfrac{-(2a-t)e^{-pt}}{p} + \dfrac{e^{-pt}}{p^2}\right]_a^{2a}}{1-e^{-2ap}}$

$= \dfrac{\left[\dfrac{-ae^{-ap}}{p} - \dfrac{e^{-ap}}{p^2} + \dfrac{e^{-0}}{p^2}\right] + \left[\dfrac{-0}{p} + \dfrac{e^{-2ap}}{p^2} + \dfrac{ae^{-ap}}{p} - \dfrac{e^{-ap}}{p^2}\right]}{1-e^{-2ap}}$

$= \dfrac{\cancel{\dfrac{-ae^{-ap}}{p}} - \dfrac{e^{-ap}}{p^2} + \dfrac{1}{p^2} + \dfrac{e^{-2ap}}{p^2} + \cancel{\dfrac{ae^{-ap}}{p}} - \dfrac{e^{-ap}}{p^2}}{1-e^{-2ap}}$

$= \dfrac{e^{-2ap} - 2e^{-ap} + 1}{p^2(1-e^{-2ap})}$

$= \dfrac{(e^{-ap}-1)^2}{p^2(1-e^{-ap})(1+e^{-ap})}$

$= \dfrac{(1-e^{-ap})^2}{p^2\cancel{(1-e^{-ap})}(1+e^{-ap})}$ $\quad [(-1)^2 = +1]$

$= \dfrac{(1-e^{-ap})}{p^2(1+e^{-ap})}$

$= \dfrac{e^{ap/2}(1-e^{-ap})}{p^2 \cdot e^{ap/2}(1+e^{-ap})}$

$= \dfrac{e^{ap/2} - e^{-ap/2}}{p^2(e^{ap/2} + e^{-ap/2})}$

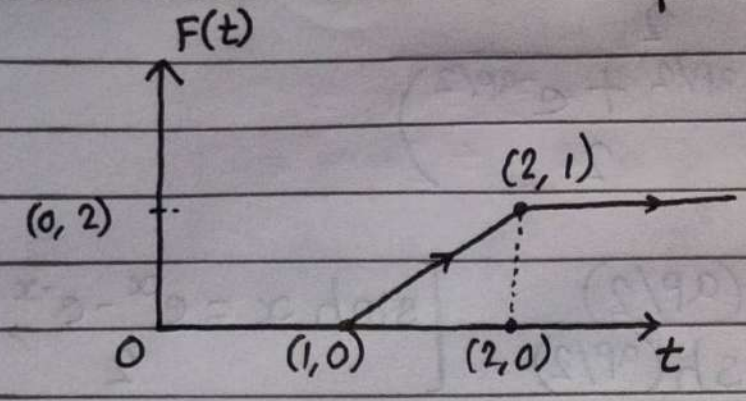
$$= \frac{\frac{e^{ap/2} - e^{-ap/2}}{2}}{p^2\left(\frac{e^{ap/2} + e^{-ap/2}}{2}\right)}$$

$$= \frac{\sinh(ap/2)}{p^2 \cdot \cosh(ap/2)} \quad \left[\sinh x = \frac{e^x - e^{-x}}{2},\ \cosh x = \frac{e^x + e^{-x}}{2}\right]$$

$$= \frac{\tanh(ap/2)}{p^2} \quad \left[\frac{\sinh x}{\cosh x} = \tanh x\right]$$

$$\therefore L(f(t)) = \frac{\tanh(ap/2)}{p^2}.$$

Q.4) Express the following in terms of unit step function and find its laplace transform.



Soln:

$$F(t) = \begin{cases} 0, & 0 \le t < 1 \\ t-1, & 1 \le t < 2 \\ 2, & 2 \le t \end{cases}$$

$$F(t) = 0.[1 - U(t-1)] + (t-1)[U(t-1) - U(t-2)] + 2[U(t-2)]$$

$$\Rightarrow F(t) = (t-1)\,U(t-1) - (t-1)\,U(t-2) + 2\,U(t-2)$$

$$\Rightarrow F(t) = (t-1)\,U(t-1) - [t-1-2]\,U(t-2)$$

$$\Rightarrow F(t) = (t-1)\,U(t-1) - (t-3)\,U(t-2)$$

We have, from second shifting theorem,

$$L(F(t-a)\,U(t-a)) = e^{-ap} f(p)$$

$$f(p) = L(F(t))$$

$$L(F(t)) = L((t-1)U(t-1)) - L((t-3)U(t-2))$$

Now, $L((t-1)U(t-1)) = e^{-p} \cdot f(p)$

$$f(p) = L(t) = \frac{1}{p^2}$$

$$\Rightarrow L((t-1)U(t-1)) = \frac{e^{-p}}{p^2}$$

Also, $L((t-3)U(t-2)) = L[((t-1)-2)U(t-2)]$

$$= e^{-2p} f(p)$$

$$f(p) = L(t-1) = L(t) - L(1) = \frac{1}{p^2} - \frac{1}{p}$$

$$\Rightarrow L((t-3)U(t-2)) = \frac{e^{-2p}}{p^2} - \frac{e^{-2p}}{p}$$

$$\therefore L(F(t)) = \frac{e^{-p}}{p^2} - \frac{e^{-2p}}{p^2} + \frac{e^{-2p}}{p}.$$

Inverse Laplace Transform

$$L(F(t)) = f(p)$$

$$\Rightarrow L^{-1}(f(p)) = F(t)$$

$f(p)$	$L^{-1}(f(p))$
$\frac{1}{p}$	1
$\frac{1}{p^2}$	t
$\frac{1}{p^{n+1}}$	$\frac{t^n}{n!}$, n is a +ve integer
$\frac{1}{p^{n+1}}$	$\frac{t^n}{\sqrt{n+1}}$, $n > -1$
$\frac{1}{p-a}$	e^{at}
$\frac{1}{p+a}$	e^{-at}

Linearly Property

$$L^{-1}(C_1 f_1(p) + C_2 f_2(p)) = C_1 L^{-1}(f_1(p)) + C_2 L^{-1}(f_2(p))$$

First Translation Property

If $L^{-1}(f(p)) = F(t)$ then

$$L^{-1}(f(p-a)) = e^{at} F(t)$$

Second Shifting Property

If $L^{-1}(f(p)) = F(t)$ then

$$L^{-1}(e^{-ap} f(p)) = G(t)$$

where, $G(t) = \begin{cases} F(t-a) & , t \geq a \\ 0 & , t < a \end{cases}$

Change of Scale Property

If $L^{-1}(f(p)) = F(t)$ then

$$L^{-1}(f(ap)) = \frac{1}{a} F\left(\frac{t}{a}\right).$$

Q.1) Evaluate: $L^{-1}\left(\frac{e^{-p} - 3e^{-3p}}{p^2}\right)$.

Soln:

$$L^{-1}\left(\frac{e^{-p} - 3e^{-3p}}{p^2}\right)$$

$$= L^{-1}\left(\frac{e^{-p}}{p^2}\right) - 3L^{-1}\left(\frac{e^{-3p}}{p^2}\right)$$

We have,

$$L^{-1}(e^{-ap} f(p)) = G(t) = \begin{cases} F(t-a), & t \geq a \\ 0, & t < a \end{cases}$$

For, $L^{-1}\left(\frac{e^{-p}}{p^2}\right) = G_1(t) = \begin{cases} F(t-1), & t \geq 1 \\ 0, & t < 1 \end{cases}$

For, $L^{-1}\left(\frac{e^{-3p}}{p^2}\right) = G_2(t) = \begin{cases} F(t-3), & t \geq 3 \\ 0, & t < 3 \end{cases}$

So,

$$L^{-1}\left(\frac{e^{-p} - 3e^{-3p}}{p^2}\right) = G_1(t) - 3G_2(t)$$

$$F(t) = \begin{cases} h(t), & t \geq a \\ h_0(t), & t < a \end{cases} \qquad h(t)[U(t-a)] + h(t)[1 - U(t-a)]$$

$$G_1(t) = F(t-1)[U(t-1)]$$
$$G_2(t) = F(t-3)[U(t-3)]$$

$$L^{-1}\left(\frac{e^{-p} - 3e^{-3p}}{p^2}\right) = F(t-1)U(t-1) - F(t-3)U(t-3)$$

$$F(t) = L^{-1}(f(p)) = L^{-1}\left(\frac{1}{p^2}\right) = t$$

$$L^{-1}\left(\frac{e^{-p} - 3e^{-3p}}{p^2}\right) = (t-1)U(t-1) - 3(t-3)U(t-3).$$

Q.2) Evaluate: $L^{-1}\left(\frac{p e^{-ap}}{p^2-\omega^2}\right)$, $a>0$.

Soln:

$$f(p)=\frac{p}{p^2-\omega^2}$$

$$L^{-1}(f(p)) = L^{-1}\left(\frac{p}{p^2-\omega^2}\right) = \cosh(\omega t) = F(t)$$

$$\Rightarrow L^{-1}\left(\frac{p e^{-ap}}{p^2-\omega^2}\right) = G(t) = \begin{cases} F(t-a), & t \geq a \\ 0, & t < a \end{cases}$$

$$= F(t-a)\,U(t-a)$$

$$= \cosh[\omega(t-a)]\,U(t-a).$$

Inverse of Laplace Transform of Derivative

If $L^{-1}(f(p)) = F(t)$ then

$L^{-1}(f^n(p)) = (-1)^n t^n F(t)$.

Multiplication by p

If $L^{-1}(F(t)) = f(p)$ and $F(0) = 0$ then

$$L^{-1}(p f(p)) = F'(t).$$

Division by p

If $L^{-1}(f(p)) = F(t)$ then

$$L^{-1}\left(\frac{f(p)}{p}\right) = \int_0^t F(t)\, dt.$$

Imp.
Convolution Theorem

If $L^{-1}(f(p)) = F(t)$ and $L^{-1}(g(p)) = G(t)$

then $L^{-1}(f(p) \cdot g(p)) = F * G = \int_0^t F(u) \cdot G(t-u)\, du$.

Q.1) Find the Laplace ~~trans~~ inverse of $\log\left(1+\frac{1}{p^2}\right)$.

Soln:

Let, $f(p) = \log\left(1+\frac{1}{p^2}\right)$

$$\Rightarrow f'(p) = \frac{-2}{p^3} \cdot \frac{1}{1+1/p^2} = \frac{-2}{p^3} \cdot \frac{p^2}{p^2+1}$$

$$\Rightarrow f'(p) = \frac{-2}{p(p^2+1)}$$

Now,

$$L^{-1}(f(p)) = F(t)$$

$$\Rightarrow L^{-1}(f'(p)) = -t\,F(t)$$

$$\Rightarrow L^{-1}\left(\frac{-2}{p(p^2+1)}\right) = -2 \cdot L^{-1}\left(\frac{1}{p} \cdot \frac{1}{(p^2+1)}\right)$$

Let, $f_1(p) = \frac{1}{p}$

$$f_2(p) = \frac{1}{p^2+1}$$

$$L^{-1}\left(\frac{1}{p} \cdot \frac{1}{(p^2+1)}\right) = \int_0^t f_1(u) \cdot f_2(t-u)\,du$$

$$F_1(t) = L^{-1}(f_1(p)) = L^{-1}\left(\frac{1}{p}\right) = 1$$

$$\Rightarrow f_1(u) = 1$$

$$F_2(t) = L^{-1}(f_2(p)) = L^{-1}\left(\frac{1}{p^2+1}\right) = \sin t$$

$$\Rightarrow f_2(t-u) = \sin(t-u)$$

So,

$$L^{-1}\left(\frac{1}{p(p^2+1)}\right) = \int_0^t \sin(t-u)\, du$$

$$= \left[\cos(t-u)\right]_0^t$$

$$= (1 - \cos t)$$

$$\Rightarrow L^{-1}\left(\frac{-2}{p(p^2+1)}\right) = -2(1-\cos t)$$

$$\Rightarrow -2(1-\cos t) = -t\,F(t)$$

$$\Rightarrow F(t) = \frac{2(1-\cos t)}{t}$$

$$\Rightarrow L^{-1}(f(p)) = L^{-1}\left(\log\left(1+\frac{1}{p^2}\right)\right) = \frac{2(1-\cos t)}{t}.$$

Convolution Theorem

If $L^{-1}(f(p)) = F(t)$ and $L^{-1}(g(p)) = G(t)$, then the inverse Laplace transform of their product is given by the convolutional integral:

$$L^{-1}(f(p).g(p)) = (F*G)(t) = \int_0^t F(u).G(t-u)\,du$$

$$= \int_0^t F(t-u).G(u)\,du.$$

Q. Using Convolution Theorem to Evaluate Inverse Laplace Transforms, (i) $L^{-1}\left(\frac{p}{(p^2+4)^2}\right)$.

Soln:

$$L^{-1}\left(\frac{p}{(p^2+4)^2}\right) = L^{-1}\left(\frac{p}{p^2+4}\cdot\frac{1}{p^2+4}\right)$$

Let, $f(p) = \frac{p}{p^2+4}$ & $g(p) = \frac{1}{p^2+4}$

We know,

$$L^{-1}(f(p).g(p)) = \int_0^t F(u).G(t-u)\,du$$

Now, $F(t) = L^{-1}\left(\frac{p}{p^2+4}\right) = \cos 2t$

$$G(t) = L^{-1}\left(\frac{1}{p^2+4}\right) = \frac{\sin 2t}{2}$$

$$2\cos A.\sin B = \sin(A+B) - \sin(A-B)$$

$$L^{-1}\left(\frac{p}{(p^2+4)^2}\right) = \frac{1}{2}\int_0^t \cos 2u \cdot \sin 2(t-u)\, du$$

$$= \frac{1}{4}\int_0^t [\sin 2t + \sin(2t-4u)]\, du$$

$$= \frac{1}{4}\left[\sin(2t)\,[u]_0^t - \left(\frac{-\cos(2t-4u)}{-4}\right)\right]_0^t$$

$$= \frac{1}{4}\left[u\sin(2t) - \frac{\cos(2t-4u)}{4}\right]_0^t$$

$$= \frac{1}{4}\cdot t\sin(2t).$$

(ii) $L^{-1}\left(\frac{p^2}{(p^2+a^2)(p^2+b^2)}\right)$

Soln:

$$L^{-1}\left(\frac{p^2}{(p^2+a^2)(p^2+b^2)}\right) = L^{-1}\left(\frac{p}{(p^2+a^2)}\cdot\frac{p}{(p^2+b^2)}\right)$$

$$= L^{-1}(f(p).g(p))$$

$F(t) = L^{-1}(f(p)) = \cos at$

$G(t) = L^{-1}(g(p)) = \cos bt$

$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$

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Page

$$L^{-1}\left(\frac{p^2}{(p^2+a^2)(p^2+b^2)}\right) = \int_0^t \cos au \cdot \cos b(t-u)\, du$$

$$= \frac{1}{2}\int_0^t \left[\cos(au+bt-bu) + \cos(au-bt+bu)\right] \cdot du$$

$$= \frac{1}{2}\int_0^t \left[\cos((a-b)u+bt) + \cos((a+b)u-bt)\right] du$$

$$= \frac{1}{2}\left[\frac{\sin((a-b)u+bt)}{a-b} + \frac{\sin((a+b)u-bt)}{a+b}\right]_0^t$$

$$= \frac{1}{2}\left[\frac{\sin(at)}{a-b} + \frac{\sin(at)}{a+b} - \frac{\sin(bt)}{a-b} - \frac{\sin(-bt)}{a+b}\right]$$

$$= \frac{1}{2}\left[\frac{\sin(at)-\sin(bt)}{a-b} + \frac{\sin(at)+\sin(-bt)}{a+b}\right]$$

$$= \frac{1}{2}\left[\frac{(a+b)[\sin(at)-\sin(bt)] + (a-b)[\sin(at)+\sin(bt)]}{a^2-b^2}\right]$$

$$= \frac{1}{2}\left[\frac{a\sin(at)-b\sin(bt)+a\sin(at)-b\sin(bt)}{a^2-b^2}\right]$$

$$= \frac{1}{2}\left[\frac{2a\sin(at)-2b\sin(bt)}{a^2-b^2}\right]$$

$$= \frac{a\sin(at)-b\sin(bt)}{a^2-b^2}.$$

Solution of Ordinary Differential Equation with Constant Coefficients

Problem: Using Laplace transform, solve the following differential equation,

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t}\sin t, \text{ where } x(0)=0 \text{ \& } x'(0)=1.$$

Soln:

Given, $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t}\sin t$

$$L\left(\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x\right) = L(e^{-t}\sin t)$$

$$\Rightarrow L\left(\frac{d^2x}{dt^2}\right) + 2L\left(\frac{dx}{dt}\right) + 5L(x) = L(e^{-t}\sin t) \quad \text{---(1)}$$

~~$L(F''(t)) = p^2$~~

$$L(F''(t)) = p^2 L(F(t)) - pF(0) - F'(0)$$
$$L(F'(t)) = pL(F(t)) - F(0)$$

Also, we have, first translation property,

$$L(e^{at}F(t)) = f(p-a) \quad [f(p) = L(F(t))]$$

$$L(\sin t) = \frac{1}{p^2+1^2} = f(p)$$

$$f(p-a) = f(p+1) = \frac{1}{(p+1)^2+1} = \frac{1}{p^2+2p+2}$$

So, eqⁿ. (1) becomes,

$$[p^2 L(F(t)) - p F(0) - F'(0)] + 2[p L(F(t)) - F(0)] + 5L(x)$$

$$= \frac{1}{p^2+2p+2}$$

- Given, $x(0) = 0$ & $x'(0) = 1$

i.e. $F(0) = x(0) = 0$
& $F'(0) = x'(0) = 1$

$$[p^2 L(x(t)) - p \cdot 0 - 1] + 2[p L(x(t)) - 0] + 5 L(x(t))$$

$$= \frac{1}{p^2+2p+2}$$

$$\Rightarrow (p^2+2p+5) L(x(t)) - 1 = \frac{1}{p^2+2p+2}$$

$$\Rightarrow (p^2+2p+5) L(x(t)) = 1 + \frac{1}{p^2+2p+2}$$

$$\Rightarrow (p^2+2p+5) L(x(t)) = \frac{p^2+2p+2+1}{p^2+2p+2}$$

$$\Rightarrow L(x(t)) = \frac{p^2+2p+3}{(p^2+2p+2)(p^2+2p+5)}$$

$$\Rightarrow x(t) = L^{-1}\left(\frac{p^2+2p+3}{(p^2+2p+2)(p^2+2p+5)}\right)$$

$$\frac{p^2+2p+3}{(p^2+2p+2)(p^2+2p+5)} = \frac{Ap+B}{p^2+2p+2} + \frac{Cp+D}{p^2+2p+5}$$

$$p^2+2p+3 = (Ap+B)(p^2+2p+5) + (Cp+D)(p^2+2p+2)$$

$$p^2+2p+3 = Ap^3 + 2Ap^2 + 5Ap + Bp^2 + 2Bp + 5B + Cp^3 + 2Cp^2 + 2Cp + Dp^2 + 2Dp + 2D$$

Comparing

$$0.p^3 + 1.p^2 + 2.p + 3 = (A+C)p^3 + (2A+B+2C+D)p^2 + (5A+2B+2C+2D)p + (5B+2D)$$

Comparing, like terms,

p^3, $A+C = 0 \Rightarrow C = -A$

p^2, $2A+B+2C+D = 1$

$\Rightarrow B+D = 1 \Rightarrow D = 1-B$

p^1, $5A+2B+2C+2D = 2$

$\Rightarrow 5A+2C+2(B+D) = 2$

$\Rightarrow 5A+2C+2.1 = 2$

$\Rightarrow 5A+2(-A) = 0$

$\Rightarrow 3A = 0$

$\Rightarrow A = 0 \Rightarrow C = 0$

p°, $5B + 2D = 3$

$\Rightarrow 5B + 2(1-B) = 3$

$\Rightarrow 5B + 2 - 2B = 3$

$\Rightarrow 3B = 1$

$\Rightarrow B = \frac{1}{3} \quad \Rightarrow D = 1 - \frac{1}{3} = \frac{2}{3}$

So,

$$x(t) = L^{-1}\left(\frac{1}{3}\cdot\frac{1}{p^2+2p+2} + \frac{2}{3}\cdot\frac{1}{p^2+2p+5}\right)$$

$$= \frac{1}{3}L^{-1}\left(\frac{1}{(p+1)^2+1}\right) + \frac{2}{3}L^{-1}\left(\frac{1}{(p+1)^2+4}\right)$$

$$L^{-1}(F(p+a)) = e^{-at}\,L^{-1}(F(p))$$

$$\Rightarrow x(t) = \frac{1}{3}e^{-t}L^{-1}\left(\frac{1}{p^2+1}\right) + \frac{2}{3}e^{-t}L^{-1}\left(\frac{1}{p^2+2^2}\right)$$

$$= \frac{1}{3}e^{-t}\sin t + \frac{2}{3}e^{-t}\frac{\sin 2t}{2}$$

$$= \frac{e^{-t}\sin t}{3} + \frac{e^{-t}\sin 2t}{3}$$

$$= \frac{e^{-t}}{3}(\sin t + \sin 2t).$$